Math_170E_21W_Discussion_Logistics

Thursday, January 6, 2022 11:42 AM

Math_170...

Math 170E Discussion Section Information/Logistics

1. Instructor: Dr. Mitia Duerinckx
2. Teaching Assistant: Osman Akar
3. **Discussion Time** (all times in PST).
   Thursdays 12-12.50pm at MS 5117
   Meeting Link: https://ucla.zoom.us/j/91051914552?pwd=ZUlncTRaWUJGKzNyNjNxb2ZTQ05vdz09
   Meeting ID: 910 5191 4552, Passcode: 488799
4. **Office Hours:** You are most welcomed to attend the office hours, even when you don't have questions but you want to see other student's questions, or you just want to chat in general. I was also UCLA undergrad and I have spent almost 7 years here, so I have quite an experience of being a math major.

   I plan to hold both in person and online office hours (via zoom of course), but this is subject to change. Of course the first two weeks OHs will be via zoom.

   - **Zoom OH:** TBD on Thursday. Subject to change. Zoom link is the same as the open OH link.
   - **Open OH:** 9-10am on Tuesdays. Here is the zoom link details:
     https://ucla.zoom.us/j/97532843825?pwd=RldkOElSd25Lc0pYT3R1NTgrbDBKQT09
     Meeting ID: 975 3284 3825, Passcode: 896714
5. Email: oak@math.ucla.edu.
6. Email Policy: Please do not send me math questions via email, instead I prefer you to come to my office hours. It is quite time consuming to answer math questions via email because I cannot write mathematical symbols easily.
7. Lecture notes: I will upload my old written notes to Canvas.
8. I plan to broadcast and record the live discussion sessions, and upload the recording to the canvas. However, undeniably there will be technical errors throughout the quarter, so safest way to attend discussion is in person.
9. Useful Links (Free materials UCLA offers):
   (a) MATLAB is free for UCLA Students: https://softwarecentral.ucla.edu/matlab-getmatlab
   (b) Microsoft Office is also free: https://www.it.ucla.edu/news/microsoft-office-proplus
   (c) You can read *The Economist* magazine for free if you are on UCLA network. If not, you can use UCLA VPN. See here for more info https://www.anderson.ucla.edu/rosenfeld-library/databases/business-databases-by-name/economist. Note that being on UCLA network is also useful to download articles from a number of publishers and also from UCLA library (NOTE: for some reason this free access does not seem to be working for a couple of weeks, but I hope they will fix it soon).

1

Q: what is probability? It is a method to quantify the possibility of occurrence of events.

Ex1: Assume we have a class of 50 students

	Boys	Girls
Blue	5	10
Green	15	20

Assume the teacher chooses one student (S) at uniformly random. (meaning each student has same chance to be picked)

Q1: what is the probability that S is a boy? → P(A)

Q1: What is the probability that S is a boy? $\rightarrow P(A)$
Q2: What is the probability that S has green eyes? $\rightarrow P(B)$

Define $A = \{S \text{ is a boy}\}$, $B = \{S \text{ has green eyes}\}$

	Boy	Girl
Blue	5	10
Green	15	20

	Boy	Girl
Blue	5	10
Green	15	20

$P(A) = \frac{20}{50}$

$P(B) = \frac{35}{50}$

Q3: What is the probability that S is a green eyed boy? (S is boy "and" S has green eyes)

Question actually asking $P(A \cap B) = \frac{15}{50}$

	B	G
B	5	10
G	15	20

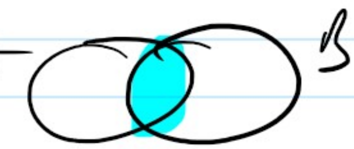
Q4: What is the prob. that either S is a boy "or" S has green eyes? $P(A \cup B) = \frac{40}{50}$

	B	G
B	5	10
G	15	20

Set Operations:

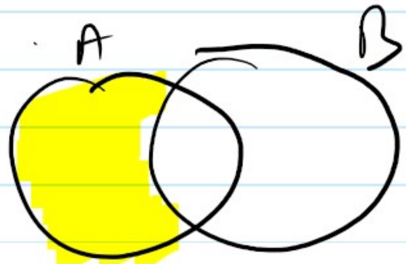
- $A \cup B = (A \text{ union } B) = A \text{ or } B$

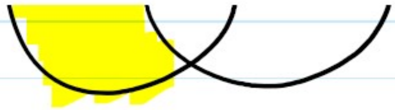
- $A \cap B = (A \text{ intersection } B) = A \text{ and } B$



- $A^c = A' = (\text{complement } A) = \text{not } A$

- $A \setminus B = A - B = (A \cap B')$





Properties: • $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

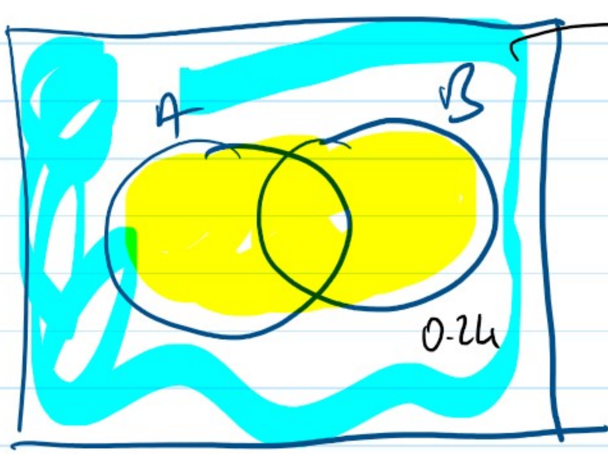
This is called "Principle of inclusion and exclusion"

• $P(A') = 1 - P(A)$

Ex1 (1.1.7) Given that $P(A \cup B) = 0.76$ and $P(A \cup B') = 0.87$, calculate $P(A)$.

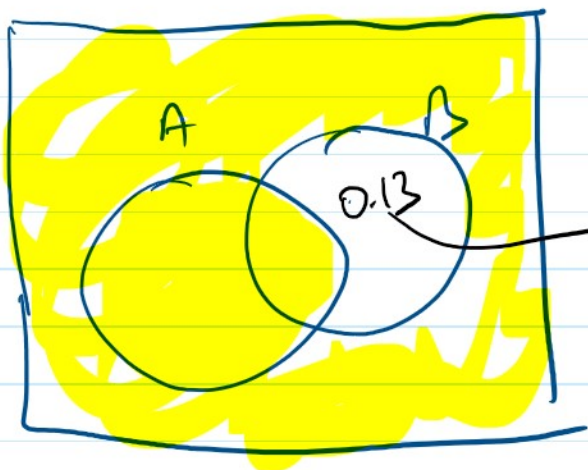
A) 0.24 B) 0.37 C) 0.5 D) 0.63 E) 0.77

Sol:



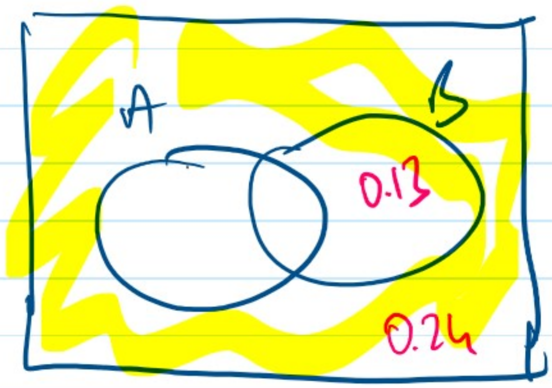
$\rightarrow (A \cup B)'$

$$P((A \cup B)') = 1 - P(A \cup B)$$
$$= 1 - 0.76$$
$$= 0.24$$



$P((A \cup B')') =$

$1 - 0.87 = 0.13$



$P(A') = 0.13 + 0.24 = 0.37$

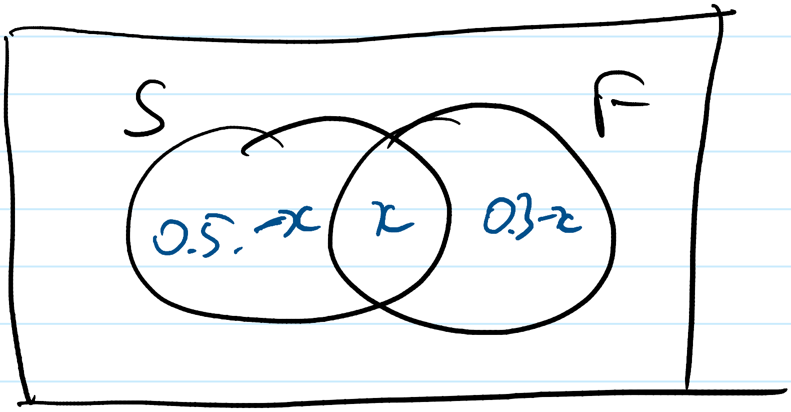
$P(A) = 1 - P(A')$
$= 1 - 0.37$
$= 0.63$

EX2: At UCLA, 70 percent of students can speak either French or Spanish. If half of the UCLA students can speak Spanish and 30% of students can speak French, what percentage of students can speak both?

A) 10 B) 20 C) 30 D) 40 E) None.

Sol: $S = \{$Student who speaks Spanish$\}$

$F = \{$Students who can speak French$\}$



$P(S \cap F) = x$? $P(F) = 0.3$, $P(S) = 0.5$

$$0.7 = P(S \cup F) = 0.5 - x + x + 0.3 - x$$

$$0.7 = 0.8 - x$$

$$x = 0.1 \to 10 \text{ percent.}$$

Sol 2: Apply PIE:

$$P(S \cup F) = P(S) + P(F) - P(S \cap F)$$

$$0.7 = 0.5 + 0.3 - P(S \cap F)$$

$$0.1 = P(S \cap F)$$

Ex 2: (1.1.1) Of a group of patients having injuries, 28% visits both physical therapist (P.T.) and a chiropractor (Ch) and 8% visits neither. Say that the prob. of visiting a P.T. exceeds the prob. of visiting Ch. by 16%. What is the prob. that a randomly selected person from this group visiting a P.T.

A) 0.24 B) 0.38 C) 0.52 D) 0.68 E) 0.72

Sol: $A = \{$ The patient visits PT $\}$

$B = \{$ The patient visits Ch $\}$

$P(A \cap B) = 0.28$, $P((A \cup B)') = 0.08$

$P(A) = P(B) + 0.16$ $\quad\quad P(A) = \underline{\underline{x}} = ?$

$\hookrightarrow P(B) = P(A) - 0.16$

$= x - 0.16$

Sol 1 (PIE).

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$1 - 0.08 = x + (x - 0.16) - 0.28$$

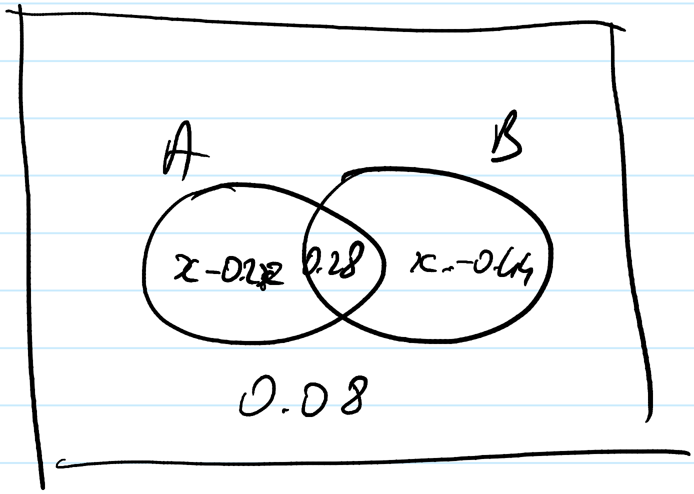
$$0.92 = 2x - 0.16 - 0.28$$

$$= 2x - 0.44$$

$$2x = 0.92 + 0.44 = 1.36$$

$x = 0.68$

sol 2



$(x - 0.28) + (0.28) + (x - 0.44) + 0.08 = 1$

$x = 0.68.$